Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **BreederNear AI**
b. Team leader name: **Sudarsan N**
c. Team member: **Shreya Azad**
d. Problem Statement: **Retail & Commerce: trusted, legal breeder-to-buyer pet commerce in India (BreederNear AI)**

Brief about the idea

“The breeder is 50 metres away. The customer never finds them.”

Why we are building this

Our founder Sudarsan N spent five years as a bird breeder, pet-shop owner and broker in Coimbatore. He started with a ₹750 setup and earned over ₹4 lakh while knowing only about 100 people. Every week he watched buyers walk past a breeder on their own street into a shop, pay more for a weaker bird, or lose money to a stranger in a WhatsApp group.

Buyers lose money and trust

- Can’t find the breeder next door, so they pay shop prices
- Can’t tell a real seller from a scam: “full advance, courier only”, stolen photos
- Don’t know which birds are protected or that dog breeders must register[5]
- Alone after buying: wrong cage, wrong food, late to the vet

Breeders lose time and reach

- Sell only inside their own small circle and district groups
- Hours of calls every day, the same 20 questions, the same post retyped in every group
- Good breeders look exactly like bad ones: no way to prove it

Animals and the law lose most

- 300+ of India’s ~1,200 bird species are found in illegal trade[3]
- Munias are dyed to disguise them[4]; dyed chicks were still on sale in an unlicensed Bengaluru shop in June 2026[6]
- Sick animals are sold because nobody checks before the sale

Still unsolved in 2026: OLX stopped live-animal listings[2], big pet platforms sell food and accessories, so India’s ~39 million-pet market[1] still buys animals through WhatsApp, Facebook and Instagram groups with no checks at all.

Founder figures are his own experience · [1] Euromonitor 2025 · [2] OLX India: items not allowed · [3] WWF-India / TRAFFIC · [4] Deccan Herald, Aug 2023 · [5] PCA (Dog Breeding and Marketing) Rules 2017 · [6] PETA India, 25 Jun 2026

From pain to product: the whole journey, end to end

Step	Today, in groups and shops	With BreederNear AI	Built and measured
Find	Walk into the nearest shop; the breeder next door is invisible	Pets near you, trusted first, a match % for your home, a fair-price bar	Match % calculated in code; 16 sample breeders, 36 sample pets
List	Breeder retypes the same post in every group	Photos + a Tamil or English message, typed or as a voice note, become a full listing	17–27 s live: upload, AI fill, trust check
Trust	No checks: stolen photos, scam prices, protected birds	Every listing screened before it goes live; Verified breeder badge	**8/8 protected species blocked, 0/37 false alarms**
Check	Pay a stranger on social media in full, in advance	Paste any outside post: “Is this post safe?”; share the warning back to the group; report protected animals	5–7 s per post; nothing stored
Talk	Endless calls and the same answers	AI drafts the buyer’s first message and the breeder’s reply; scam shield on both	Scam scan on every enquiry and reply
Buy	“Courier only”, no handover	Book a visit, see the pet, pay in person, tap “I bought this pet”	No payments in the app, by design
After	New owner on their own	Starter kit sized for welfare, 14-day plan, daily log, growth, symptom urgency	**Never a medicine or dose (evals E11, E19)**
Grow	Breeder has no data	Dashboard: views, enquiries, visits and sales, week on week	8-week charts and trend cards

One tool layer, two ways in: every step works by tapping or by asking seven Gemini agents in English or Tamil. Live now: 69/69 agent eval runs, 339 unit tests, 0 accessibility problems.

Opportunities

How is it different?

WhatsApp / Facebook groups: reach, but no checks, no records, no recourse

Classifieds (OLX): stopped live-animal listings

Pet super-apps: food, accessories and vets, not the animal

Kennel websites: one seller vouching for itself

No platform we found checks every listing for wildlife law, registration, scam prices and reused photos before it goes live, and then stays with the buyer after the sale.

How does it solve the problem?

Invisible breeders → listed in a minute from photos and a Tamil message

Scams → price, wording and reused-photo checks; outside posts checked too

Illegal birds → blocked in code from the photo alone

Unproven breeders → Verified badge: phone, facility photos, registration

Lost hours → AI drafts replies; visits booked in the app

Alone after buying → starter kit, care plan, daily log, vet urgency

USP

Trust built in, not bolted on.

- Rules in code decide; Gemini reads and explains. A prompt injection still ends in CAUTION (eval E12).
- Protected species blocked from the photo alone: 8/8 in testing, 0/37 false alarms.
- Works where the trade already happens: Tamil and English messages, posts from groups.
- Founder–market fit: designed from five years inside this exact trade.

List of features offered by the solution

For buyers (customer app)

- Pets near me with a match % for your home, trusted first, fair-price bar
- Listing page: trust checks, health notes, questions to ask
- AI-drafted first message; scam warnings on replies
- Book a visit; “I bought this pet” after it (paid in person)
- My pets: daily log with AI feedback, growth, health calendar
- Symptom checker: how urgently to see a vet (never medicines)
- “Is this post safe?” with share and report; starter kit; Local Breeders

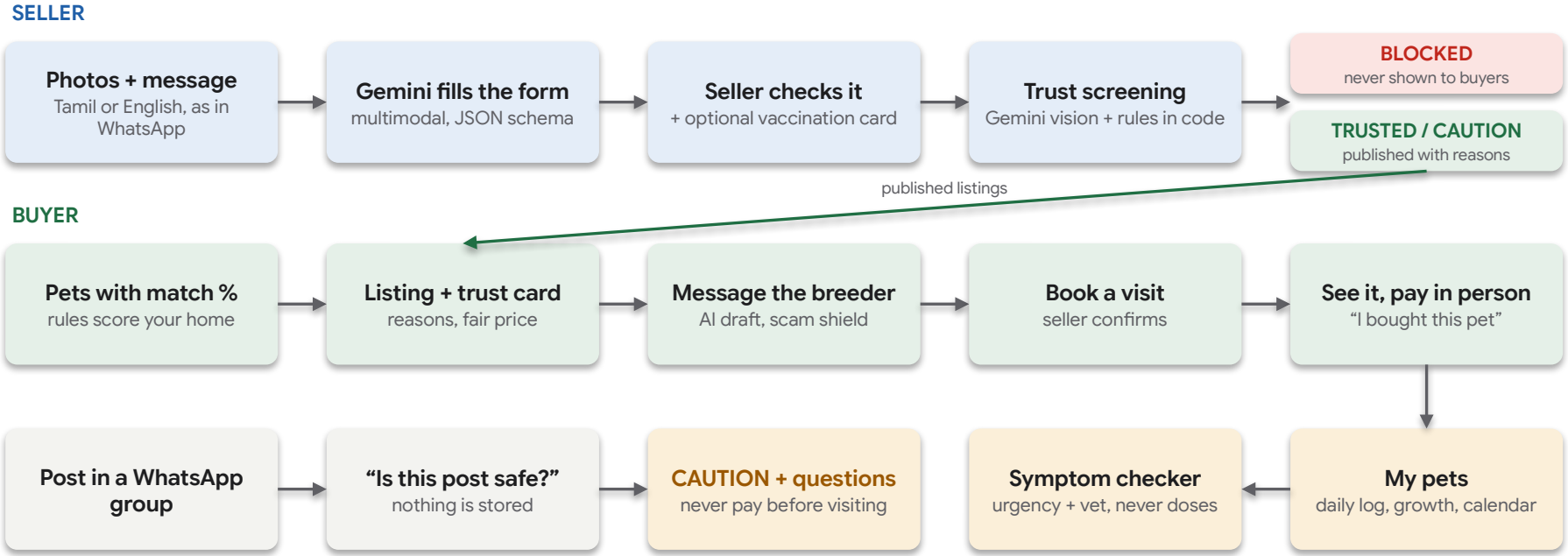
For sellers (seller app)

- “Fill with AI”: photos + a Tamil/English message or voice note become a listing
- Publish with trust check: TRUSTED, CAUTION or BLOCKED, with reasons
- Dashboard charts: views, enquiries, visits, sales, funnel
- Verified breeder badge: phone code, facility photos read by Gemini, dog-breeder certificate (no ID documents)
- Visit requests to confirm; AI-drafted replies to enquiries
- Vaccination card read into the trust check

Everywhere

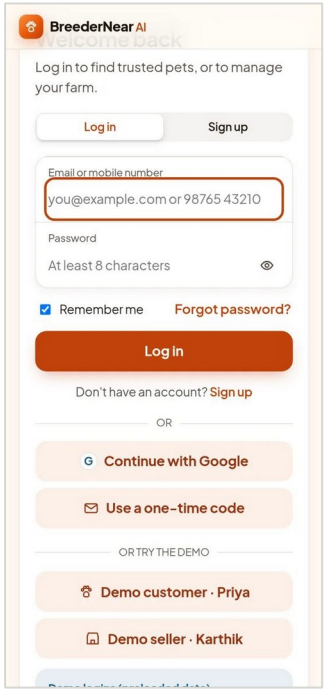
Seven Gemini agents in English or Tamil use the same tools as the screens. **Sign-in:** password, one-time code or Google (Firebase Authentication). **Safety:** roles checked in the API and the AI tools, rate limits, every AI reply checked for medicine doses. **Accessible:** WCAG 2.2 AA, phone to desktop.

Process flow diagram or Use-case diagram

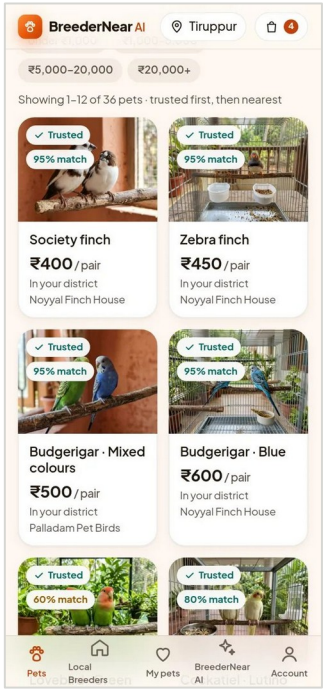


Every step works by tapping, or by asking BreederNear AI in chat: the same tools, the same checks. Sellers also earn a Verified breeder badge.

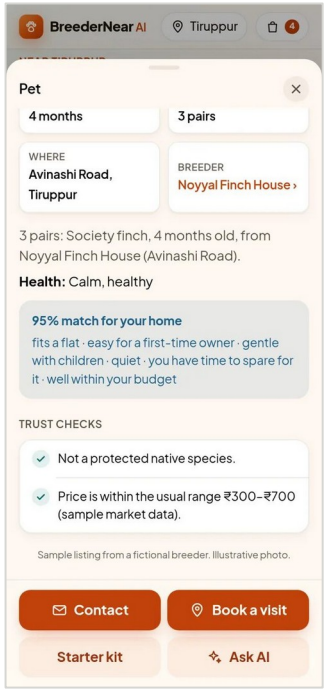
Wireframes/Mock diagrams of the proposed solution (optional)



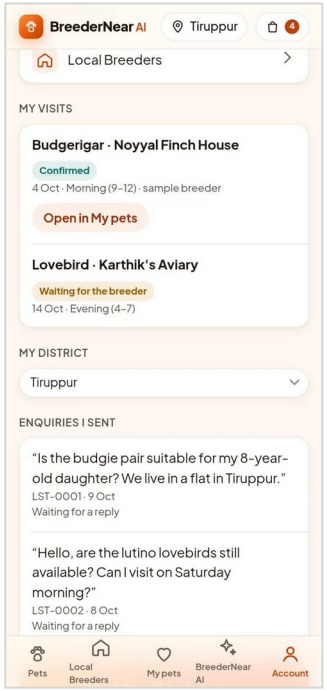
Sign in or one-tap demo



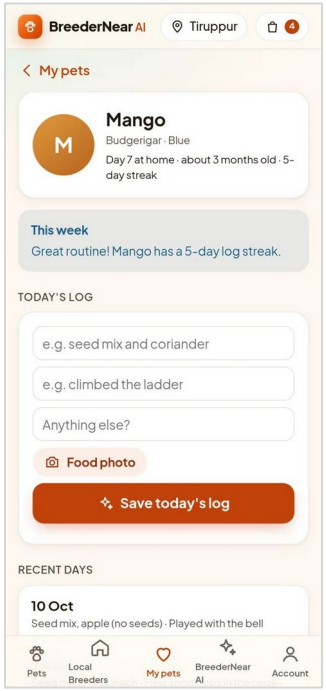
Pets with match %



Listing + trust checks

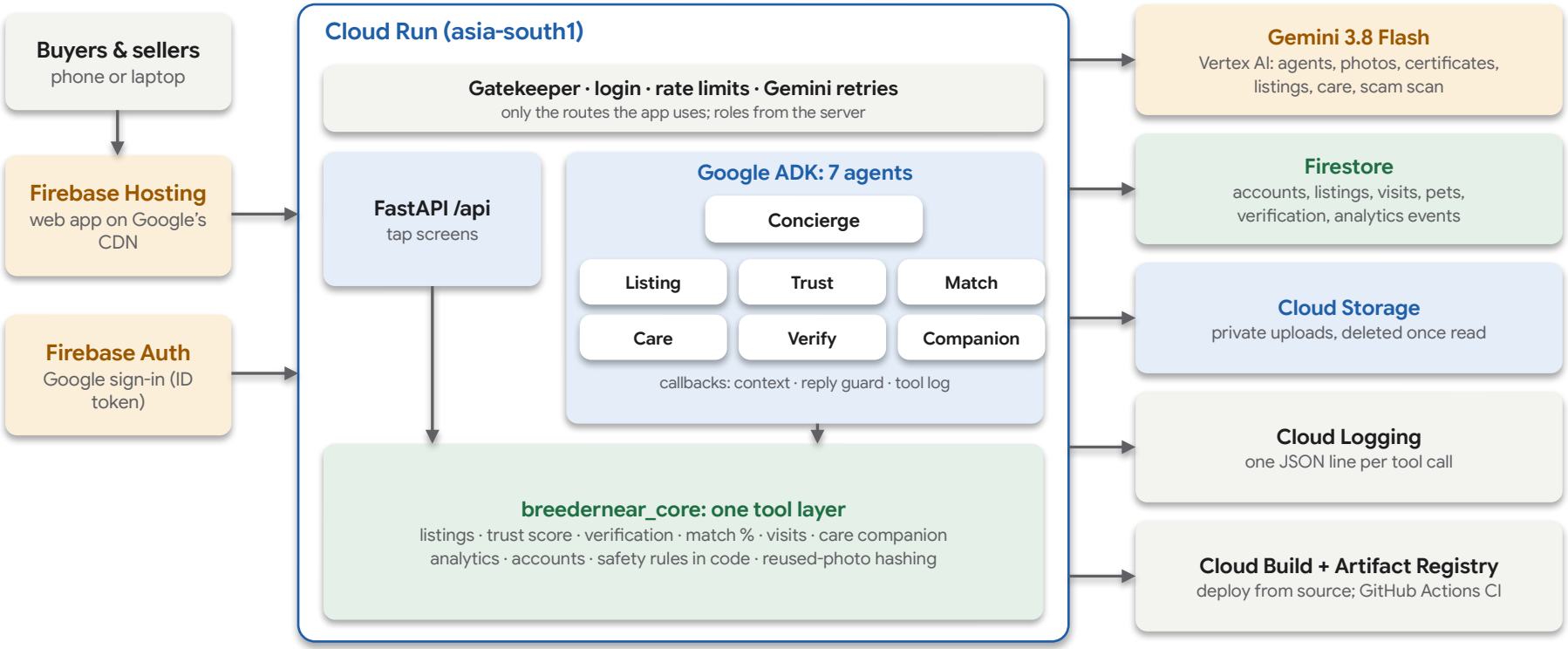


Visits, then "I bought it"



My pets companion

Architecture diagram of the proposed solution



Technologies to be used in the solution

Google technology	How BreederNear AI uses it
Gemini 3.8 Flash (Vertex AI)	Seven ADK agents; listings from photos, Tamil/English text and voice notes; photo, facility and certificate reading; care plans, daily-log feedback, symptom urgency; message drafts and scam scan. JSON schemas.
Google ADK (Python)	Concierge + listing, trust, match, care, verification and companion agents; tools; callbacks on every agent; streamed replies
Cloud Run	API and agents; login gatekeeper; rate limits; Gemini retries with backoff
Firebase Hosting	The web app on Google's CDN; calls Cloud Run directly (CORS) so AI replies stream
Firebase Authentication	Continue with Google; the server verifies the Firebase ID token
Firestore	Accounts, listings, enquiries, visits, sales, pets and logs, verification, analytics
Cloud Storage, Cloud Build, Artifact Registry, Cloud Logging	Private uploads; deploy from source; tool-call logs
gemini-3.1-flash-image	Illustrative photos for the fictional sample listings (labelled in the app)

Also used

- Python 3.12, FastAPI, Pydantic
- Pillow: our own perceptual hash (dHash) for reused photos
- Plain HTML, CSS, JavaScript and SVG charts: no build step
- pytest, Playwright + axe-core, GitHub Actions
- Firebase CLI (deploy only)

Google Cloud AI map: which AI does each job, and how well

Google Cloud AI	What it does in BreederNear AI	Measured
Gemini vision	Listing photos: species, protected birds, dye, blur, watermarks, stock photos. Facility photos, dog-breeder certificates, vaccination cards, food photos	**8/8 protected species blocked; 0/37 false alarms; 202/202 photo checks**
Gemini audio	A breeder's Tamil or English voice note becomes a full listing, with a transcript; no separate speech-to-text step	**Tamil voice note → correct listing (3/3 eval runs, E23)**
Gemini, multilingual	Romanised Tamil and English sale messages; a buyer's own words become a home profile	Tamil message → same listing (E02); profile → match % (E21)
Gemini structured output	12 task types, each with a JSON schema. Gemini reads and writes; code decides trust, badges, match % and urgency	Prompt injection still ends in CAUTION (E12)
Google ADK	7 agents (a concierge and 6 specialists) and 29 tools, the same tools the tap screens use; streamed replies	**69/69 agent eval runs passed (23 cases × 3 runs)**
ADK callbacks	Context before every model call; every reply checked for medicine names and doses, even while streaming; every tool call logged	No drug names or doses (E11, E19)
Vertex AI	Gemini 3.8 Flash on the global endpoint, retries with backoff on 408, 429 and 5xx	Listing in 17–27 s; post check 5–7 s
Firebase Auth, Cloud Run, Firestore, Storage	Google sign-in; the API and agents; all data; private uploads	Smoke test 17/17 on every release

Estimated implementation cost (optional)

Business case, scale and go-to-market

Market: India's pet care products market is USD 9.2 B in 2026, heading to USD 13.9 B by 2034[7], with about 39 million pets[1]. The animals themselves are still sold with no checks: that is our wedge.

Revenue (planned)

- Verified-breeder subscription: badge, dashboard insights, more listings
- Featured listings per district
- Starter-kit and accessory commerce after every sale (cart already built)
- B2B trust-screening API for classifieds and pet shops

Built to scale

- Trust costs ≈ ₹1.3 per listing (measured) against pets priced ₹400–₹25,000
- Serverless Cloud Run + Firestore: near zero when idle, scales out per request
- Districts, species, fair prices and protected lists are data, not code: a new district is a data file
- Gemini already reads Tamil, English and mixed messages

Go-to-market

- 1 · Pilot in Coimbatore and Tiruppur with the founder's breeder network
- 2 · Tamil Nadu, district by district; list by sending a WhatsApp message
- 3 · Other states and languages; the trust API for other marketplaces
- Buyers arrive through "Is this post safe?" links shared in the same groups

Growth loop: verified breeders → trusted listings → more buyers → more sales and care data → sharper fair prices and trust scores → more breeders join.

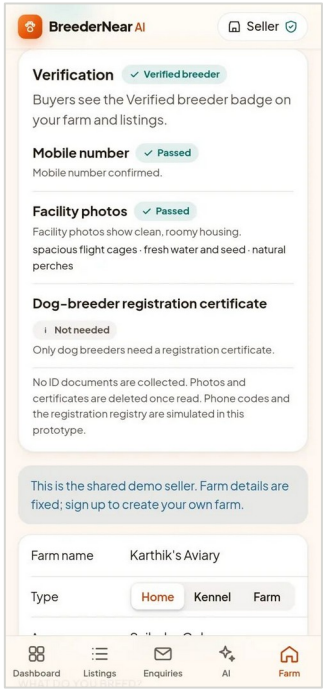
Impact we measure, live in the app: listings checked before going live · protected animals stopped · scam warnings raised · verified breeders. Every warning can be shared back to the group it came from, and protected species come with a way to report them.

Revenue lines and rollout are plans, not results · [1] Euromonitor 2025 · [7] IMARC Group, India Pet Care Products Market (2026) · Cost: docs/submission/05-deck-sources.md

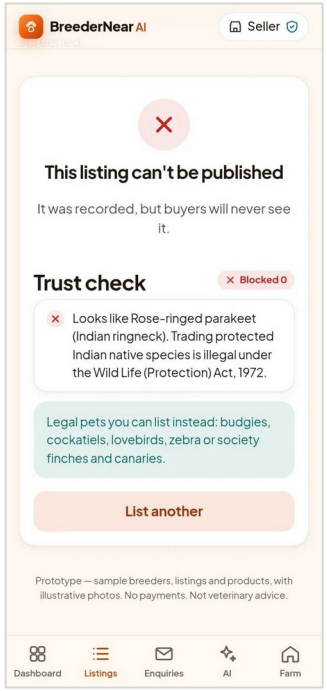
Snapshots of the prototype



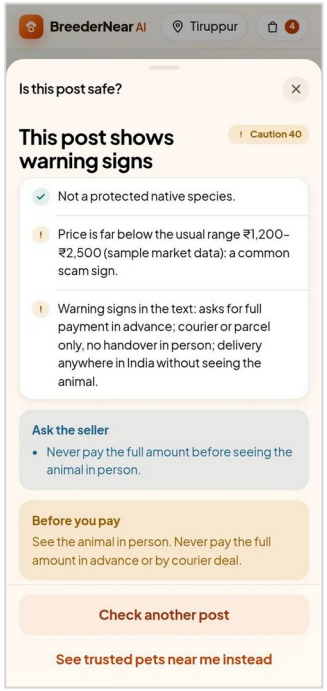
Seller analytics



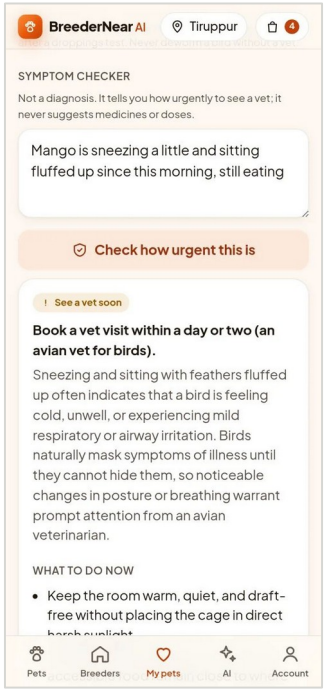
Verified breeder



Protected species: blocked



“Is this post safe?”



Symptom urgency, no doses

Prototype Performance report/Benchmarking

Quality (real Gemini, 10–12 Oct 2026)	Result
Agent evals: 23 cases × 3 runs (listing, Tamil, blocking, scams, reused photos, prompt injection, no medicine advice, roles, verification, visits, care companion)	69/69 passed
Protected species blocked from the photo alone	8/8
Ordinary pets wrongly flagged as protected	0/37
All photo checks: species, dye, blur, no animal, watermark, screenshot	202/202*
Accessibility: axe-core, WCAG 2.2 AA, light + dark, 320–1280 px, all screens	0 problems
Unit tests in CI · live smoke test	339 · 17/17

Speed (live)	Time
List a pet: upload, AI fill, publish with trust check	17–27 s
Check a pasted post	5–7 s
Web app load (Firebase CDN)	≈ 0.7 s
Gemini: read a certificate · scan a message	8–9 s · 2–3 s
Gemini: symptom urgency · daily-log feedback	7–8 s · 4–10 s

How we test

Gemini isn't deterministic, so each eval runs three times and we report the count. Decisions (trust, verification, match %, urgency floor) are made by tested code.

*Test photos are AI-generated and the screening prompt was tuned on them (first run 196/202), so treat this as an upper bound · Reproduce: scripts/run_evals.sh 3 · scripts/vision_accuracy.py · pytest -m ui tests/ui · evals/RESULTS.md, evals/VISION_RESULTS.md

Additional Details/Future Development (if any)

Roadmap (not in the prototype)

- Pilot with real breeders in Coimbatore and Tiruppur, from the founder's network
- Real SMS and email delivery for one-time codes
- List by sending a WhatsApp message (WhatsApp Business Platform)
- Live State Animal Welfare Board and PARIVESH registry checks; the platform registrations India's pet-trade rules require
- Chat history in Agent Engine sessions, so Cloud Run can scale past one instance
- Breeder ratings from buyers' care-companion outcomes; vet and pet-pharmacy partners; more languages

Responsible by design (already built)

- Protected species blocked in code, whatever the model says
- No medicine names or doses: checked on every AI reply, even while streaming
- Verification without ID documents or selfies; photos deleted once read
- Codes shown on screen only open new or demo accounts, never someone else's
- Symptom checker: red flags are always an emergency; never a diagnosis
- Synthetic sample data, clearly labelled; simulated registry numbers start with SIM-

Provide links to your:

- **GitHub Public Repository**: https://github.com/sudarsan-22/breedernear-ai
- **Demo Video Link (3 Minutes)**: [add the YouTube link on 17 Oct]
- **Final Product Link**: https://breedernear-ai-2026.web.app (API: https://breedernear-655711985039.asia-south1.run.app)

Try it in 2 minutes: demo accounts with preloaded data

Customer: priya.customer@example.com **Seller:** karthik.seller@example.com **Password:** demo12345

Or tap “Demo customer · Priya” / “Demo seller · Karthik” on the login screen. The README has a step-by-step walkthrough.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

